POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the "Rational Thinking" as an aspect of Artificial Intelligence. List out the various application areas of Artificial Intelligence. 8

   b) How can an agent be qualified as intelligent? Discuss various types of intelligent agents. 7

2. a) Solve the following Crypto arithmetic problems. 8

$$\begin{array}{r} \text{TWO} \\ +\ \text{TWO} \\ \hline \text{FOUR} \end{array}$$

   b) What do you mean by Heuristics? Compare the Greedy best first search and A* search in terms of their completeness with suitable example. 7

3. a) The game of NIM is played as follows: 8
   Two players alternatively remove one, two or three coin from a stack initially containing five coin. The player who picks up the last coin loses.

      i. Draw the full game tree.
      ii. Show that the player who has the second move can always win.
      iii. Execute $\alpha - \beta$ procedure on the game tree. How many terminal nodes are examined?

   b) How can you say that the predicate logic is more expressive than propositional logic? Explain with examples. 7

4. a) Why do we need to represent knowledge? Convert following sentences into FOL sentences. 8

      i. Everyone loves Ram.

सुगम स्टेसनरी सप्लायर्स एण्ड फोटोकपी सर्भिस
बालकुमारी, ललितपुर ९८४९५९९५९२
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ii. Not everyone loves Ravana.
iii. People protest the politicians they dislike.
iv. Politicians can fool some of the people all of the time, all of the people some of the time.
v. Not all citizens are loyal to their employers.

b) When is it useful to use clustering? The following data set contains the scores of two variables on each of seven individuals. Taking 1 and 4 as initial cluster center, apply k-means clustering to group the given data.

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

5. a) What is machine learning? Explain the learning approach used by Naive Bayes. How is it different from the approach used by neural network?

b) What do you mean by supervised learning? Explain how support vector machine (SVM) is useful for the classification in non-linearly separable data space.

6. a) Why is Natural Language understanding is difficult? Explain about Morphological Analysis with relevant example.

b) Explain the basic components of expert system. How can we make expert system knowledge base reusable?

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7. Write short notes on: (**Any two**)
a) Inference and reasoning
b) Deep learning
c) Adaline network.

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